NLP Coursework

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1 Exercise 1: Critical Paper Review

Paper reviewed: Pérez-Almendros et al., "Don't Patronize Me! An Annotated Dataset with Patronizing and Condescending Language towards Vulnerable Communities", COLING 2020.

Q1. Primary Contributions

The paper makes three main contributions.

A new annotated dataset. The central contribution is the *Don't Patronize Me!* (DPM) dataset: 10,637 paragraphs drawn from the News on Web (NoW) corpus, spanning 20 English-speaking countries and 10 keywords associated with vulnerable communities (e.g. *refugee*, *homeless*, *women*). Each paragraph is annotated for the presence of Patronising and Condescending Language (PCL) towards vulnerable groups. Prior to this work, no large-scale dataset existed specifically targeting PCL in media discourse; related corpora (e.g. the Talkdown corpus of Wang and Potts, 2019) focused on direct social-media interactions rather than journalistic language.

A taxonomy of PCL categories. The authors introduce a two-level taxonomy of seven fine-grained PCL categories grouped under three higher-level types: *The Saviour* (Unbalanced power relations, Shallow solution), *The Expert* (Presupposition, Authority voice), and *The Poet* (Metaphor, Compassion, The poorer, the merrier). This taxonomy is grounded in sociolinguistic theory and operationalised for NLP annotation, enabling fine-grained, span-level labelling that goes beyond binary PCL detection.

Baseline experiments. The paper provides a systematic empirical evaluation across a range of models — from TF-IDF SVMs and Word2Vec-based BiLSTMs to fine-tuned BERT, RoBERTa, and DistilBERT — on both binary PCL detection (Task 1) and multi-label category classification (Task 2). These baselines establish a performance floor and surface which PCL categories are hardest for current NLP systems.

Q2. Technical Strengths

Careful, multi-stage annotation design. Rather than asking annotators for a simple binary label, the authors use a three-way per-annotator scale (0 = no PCL, 1 = borderline, 2 = clear PCL), then combine the two primary annotators' judgements into a 5-point composite score and route all total disagreements (0 vs. 2) to a third referee annotator. This design captures genuine annotation uncertainty and produces a more nuanced label distribution, avoiding the artefacts that arise when borderline cases are forced into hard categories. Reporting Inter-Annotator Agreement both with and without borderline cases (41% vs. 61% Cohen's Kappa) honestly conveys both the difficulty of the task and the achievable upper bound.

Thorough and transparent error analysis. Tables 5 and 6 present concrete misclassified examples with gold and predicted labels. The analysis explains *why* the model fails in each case — for instance, that flowery adjectives associated with the Compassion category can appear in non-condescending political writing, causing false positives, or that detecting Presupposition requires external world knowledge unavailable to the model. This level of qualitative insight goes beyond reporting aggregate metrics and makes the paper genuinely useful for researchers designing future models.

Breadth of baseline models. By including models spanning the full complexity spectrum — random guessing, BoW SVM, word-embedding SVM, BiLSTM, and four BERT-family variants — the paper gives a clear picture of where complexity pays off and where it does not (notably, BERT-large underperforms BERT-base, motivating discussion of overfitting on a small dataset). This makes it easy for follow-up work to position itself relative to meaningful comparisons.

Q3. Key Weaknesses

Class imbalance is acknowledged but not addressed experimentally. The dataset contains roughly 995 positive examples out of 10,637 (approximately 9.3%), yet the experimental section makes no attempt to handle this severe imbalance — no oversampling, undersampling, or class-weighted loss is applied. The results are reported only in terms of the positive-class F1, precision, and recall, without discussing how the

threshold is chosen or how sensitive results are to it. Given that this imbalance is likely to be one of the main practical challenges for downstream users of the dataset, the omission is a significant gap.

Insufficient justification for the binary label threshold. The paper collapses the 5-point composite label (0–4) into a binary signal by treating labels 0–1 as negative and 2–4 as positive. This threshold is chosen without justification. Labels 0 (both annotators say no PCL) and 4 (both say clear PCL) are the only truly clean cases; grouping label 2 (both borderline) with label 4 (both clearly positive) inflates the training signal for ambiguous examples. The authors do not include an ablation comparing different threshold choices, leaving it unclear how sensitive results are to this decision.

BERT-large underperformance is not explained with evidence. The paper notes, correctly, that BERT-large performs worse than BERT-base and BiLSTM on Task 1, and attributes this to overfitting. However, no learning curves, training/validation loss plots, or any quantitative evidence for overfitting are provided. Without this, the explanation remains speculative. It is equally plausible that the large model simply requires different hyperparameters (e.g. a lower learning rate or more warmup steps), and the authors do not explore this.

2 Exercise 2: Exploratory Data Analysis

The following two EDA techniques are drawn directly from the appendix's NLP EDA workflow. Together they address all three of the appendix's stated reasons why EDA is critical: detecting class imbalance, determining an appropriate `max_length`, and assessing whether the task can be solved by simple keyword patterns.

Technique 1: Class Distribution and Token Length Distribution

(Appendix: Basic Statistical Profiling)

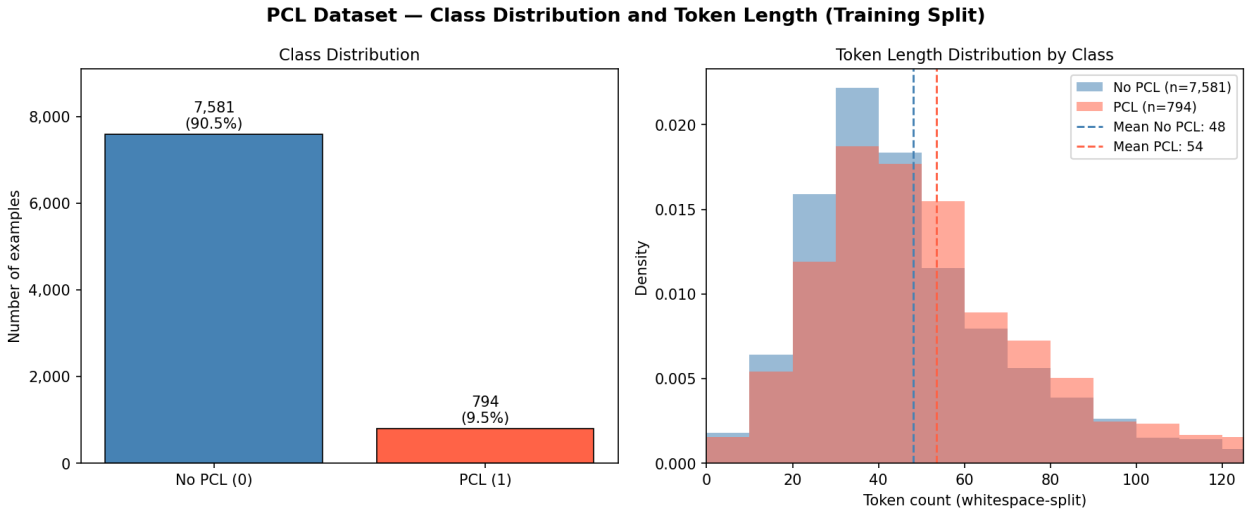

Figure 1: Left: class distribution of the training split. Right: overlapping density-normalised histograms of whitespace-tokenised text lengths for each class, with dashed vertical lines at each class mean.

Analysis. Figure 1 reveals two structural properties of the training data. First, the dataset is severely class-imbalanced: 7,581 examples (90.5%) carry the No PCL label (0) and only 794 (9.5%) carry the PCL label (1), a ratio of approximately 9.55:1. Second, PCL examples are marginally longer than non-PCL examples on average (mean 53.5 vs. 48.2 whitespace tokens; median 47 vs. 42; 95th percentile 114 vs. 101 tokens), though the distributions overlap substantially.

Impact Statement. The severe class imbalance means that a model trained with a standard cross-entropy loss and no correction will converge towards predicting the majority class, achieving ∼90% accuracy while

failing almost entirely on PCL examples — precisely the behaviour that makes F1 on the positive class the appropriate metric rather than accuracy. This motivates applying a **class-weighted loss** (weighting the PCL class by the inverse frequency ratio, ≈9.55) or oversampling minority examples during training. The token-length distribution, with a 95th percentile of 114 tokens, informs the choice of `max_length` = **128** for the transformer tokeniser: this covers the vast majority of both classes without wasting memory on excessive padding.

Technique 2: N-gram Analysis — Top Bigrams by Class

(Appendix: Lexical Analysis)

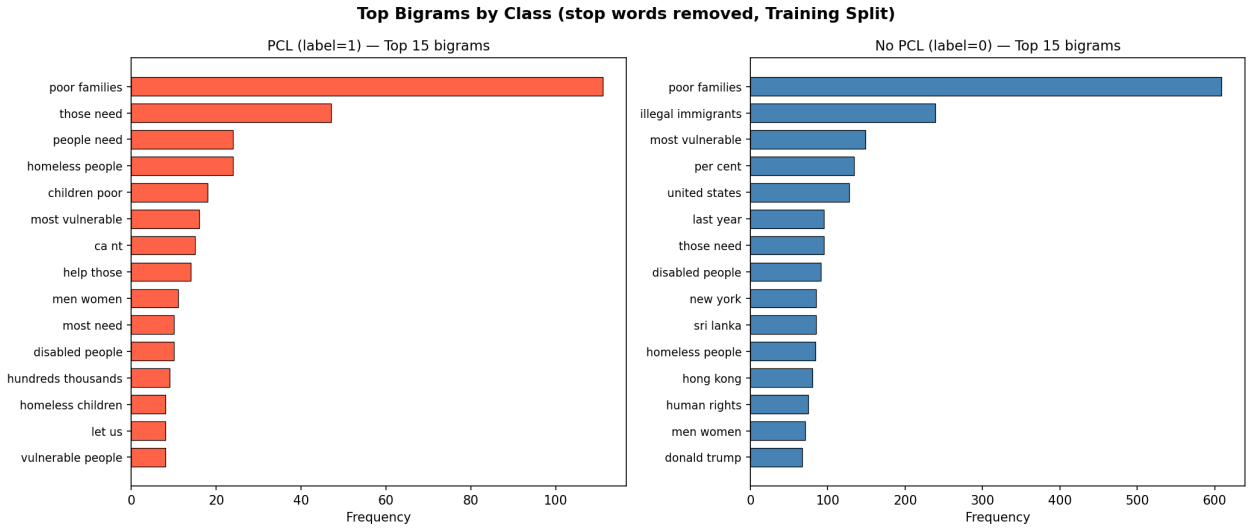

Figure 2: Top 15 most frequent bigrams (stop words removed) in PCL examples (left) and No PCL examples (right) of the training split.

Analysis. Figure 2 compares the most frequent bigrams within each class after filtering English stop words. While both classes share *poor families* as their top bigram, their subsequent distributions differ qualitatively. The PCL class is dominated by **relational, needs-oriented phrases**: *those need*, *people need*, *homeless people*, *children poor*, *help those* — language that positions the vulnerable group as passive recipients of concern or charity. The No PCL class, by contrast, skews toward **neutral, factual, and geographic constructions**: *illegal immigrants*, *per cent*, *united states*, *last year*, *new york*, *sri lanka* — consistent with straightforward news reporting about these communities without condescending framing.

Impact Statement. This analysis confirms that the task is *not* trivially solvable by topic keywords alone: both classes discuss the same vulnerable groups using many of the same content words. The distinction lies in the **framing and stance** of the language — condescending texts adopt a "helper" perspective, while neutral texts report facts. This rules out simple bag-of-words models as a viable approach and directly motivates using a **contextual language model** (such as RoBERTa) that can capture sentence-level framing and syntactic relationships rather than word co-occurrence counts. It also suggests that hand-crafted features based on topic alone (e.g. presence of the word *homeless*) would be unreliable, since such words appear frequently in both classes.

3 Exercise 3: Proposed Approach

Proposed Approach

The proposed model is a **multi-task RoBERTa-base** with two classification heads — a primary binary PCL head trained with **focal loss**, and an auxiliary 7-category head trained with **BCE loss** — jointly

optimised on the full training set, with a **decision threshold optimised on the dev set**.

Architecture

The backbone is `roberta-base` (HuggingFace Transformers), a 125M-parameter encoder pre-trained with masked language modelling on a large English corpus using dynamic masking and no next-sentence prediction objective. RoBERTa-base produces a 768-dimensional `[CLS]` representation well-suited to sentence-level classification tasks.

The `[CLS]` representation is passed through a dropout layer ($p = 0.1$) and fed into **two parallel linear heads**:

- **Binary head** ($768 \rightarrow 2$): predicts PCL vs. No-PCL. This is the primary classification head used at inference time.
- **Category head** ($768 \rightarrow 7$): predicts which of the 7 fine-grained PCL categories are present (multi-label). This head is used *only during training* as an auxiliary regulariser.

The combined loss is: $\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda \cdot \mathcal{L}_{\text{BCE}}$, where $\lambda = 0.5$.

Training Methodology

Five key deviations from the baseline are made, each justified by the EDA findings:

1. **Focal loss (replacing class-weighted cross-entropy).** The EDA revealed a 9.55:1 class imbalance. Rather than applying static class weights, focal loss [**?**] dynamically down-weights well-classified examples: $\text{FL}(p_t) = -\alpha_t(1-p_t)^\gamma \log(p_t)$, with $\gamma = 2.0$ and $\alpha = [1.0, 9.55]$. This focuses the model's learning on *hard* examples — ambiguous cases near the decision boundary — rather than the easy-to-classify majority.
2. **Multi-task learning with auxiliary category prediction.** The dataset provides 7 fine-grained PCL category labels (Unbalanced power relations, Shallow solution, Presupposition, Authority voice, Metaphor, Compassion, The poorer the merrier) for every example. By training the shared encoder to simultaneously predict these categories via a BCE loss head, the model is forced to learn richer, more structured representations of PCL. At inference time, only the binary head is used; the category head serves purely as a training-time regulariser that improves generalisation.
3. **More training epochs.** The baseline trains for only 1 epoch. The model is trained for **5 epochs** with the best checkpoint (by dev F1 of the positive class) saved.
4. **Learning-rate schedule with warmup.** A linear warmup over 10% of total training steps followed by linear decay is applied. This stabilises early training by preventing large gradient updates before the model has developed useful representations. The baseline uses no scheduler.
5. **Decision threshold optimisation.** After training, the threshold τ is swept over $\{0.05, 0.10, \ldots, 0.95\}$ on the dev set and the value maximising positive-class F1 is selected. This directly optimises the evaluation metric.

The full hyperparameter configuration is listed in Table 1.

Rationale and Expected Outcome

Why RoBERTa-base? RoBERTa-base is a well-established encoder for sentence classification tasks. It uses the same hidden size (768) and architecture as the BERT baseline family, providing a strong and well-understood foundation. It was pre-trained on a substantially larger corpus than BERT with improved training methodology (dynamic masking, no NSP objective, larger batches), giving it better general language representations while remaining computationally tractable for fine-tuning on a dataset of this size.

Why focal loss over class-weighted cross-entropy? Static class weights treat all minority-class examples equally, regardless of difficulty. Focal loss introduces a *dynamic* modulating factor $(1-p_t)^\gamma$ that automatically

Hyperparameter	Value
Backbone	`roberta-base`
Max sequence length	128
Primary loss	Focal loss, $\gamma = 2.0$, $\alpha = [1.0,\ 9.55]$
Auxiliary loss	BCEWithLogitsLoss (7 PCL categories)
Auxiliary weight (λ)	0.5
Training epochs	5
Batch size	16
Learning rate	2×10^{-5}
Warmup	10% of total steps (linear)
LR schedule	Linear decay after warmup
Optimiser	AdamW, weight_decay = 0.01
Training examples	8,375 (full set, no downsampling)

Table 1: Hyperparameter configuration for the proposed multi-task RoBERTa-base model.

reduces the contribution of easy examples and focuses training on hard, ambiguous cases near the decision boundary. This is particularly relevant for PCL detection, where the boundary between condescending framing and neutral reporting is often subtle.

Why multi-task learning with category prediction? The 7 fine-grained PCL categories capture distinct linguistic strategies (e.g. "Unbalanced power relations" vs. "Shallow solution" vs. "Metaphor"). Training the shared encoder to recognise these categories forces it to learn structured, category-aware representations rather than collapsing PCL into a single undifferentiated class. This acts as an inductive bias: the encoder must disentangle the *mechanisms* of condescension, which improves generalisation on the binary task. The auxiliary head is discarded at inference, adding zero computational cost.

Why threshold optimisation? For a model trained with focal loss and class-specific α weights, the output probability distribution is not calibrated for a fixed 0.5 threshold. Grid-searching τ on the dev set directly targets the evaluation criterion used by the shared task.

Expected outcome. Each component addresses a specific failure mode of the baseline. Focal loss and multi-task learning improve hard-example handling and representation quality respectively, while threshold optimisation directly targets the evaluation metric. A combined dev F1 in the range 0.55–0.65 is the projected outcome, compared to the baseline F1 of 0.48.

4 Exercise 5: Predictions and Error Analysis

Exercise 5.1 — Dev and Test Predictions

Dev predictions (`dev.txt`) and test predictions (`test.txt`) were generated by the multi-task RoBERTa-base model described in Exercise 3, with a decision threshold of $\tau = 0.55$ selected by grid search on the dev set. The resulting dev F1 (positive class) is **0.6196**.

Exercise 5.2 — Local Evaluation

Part A: Error Analysis

The best model (dev F1 = 0.62) and the spec baseline (single-task RoBERTa-base, 1 epoch, standard cross-entropy, 2:1 negative downsampling, $\tau = 0.5$; dev F1 = 0.48) were compared on all 2,094 dev examples. Each example was assigned to one of four mutually exclusive outcomes depending on which system, if either, predicted the gold label correctly.

The best model improves on the baseline in 199 cases while regressing on only 43 (a net gain of 156 examples), confirming that the overall F1 improvement is genuine and broadly distributed rather than concentrated on

Scenario	Count	%
Both correct	1744	83.3
Both wrong	108	5.2
Only best model correct	199	9.5
Only baseline correct	43	2.1
Total	2094	100.0

Table 2: Four-bucket comparison of the best model and the baseline on the 2,094 dev examples.

a few examples.

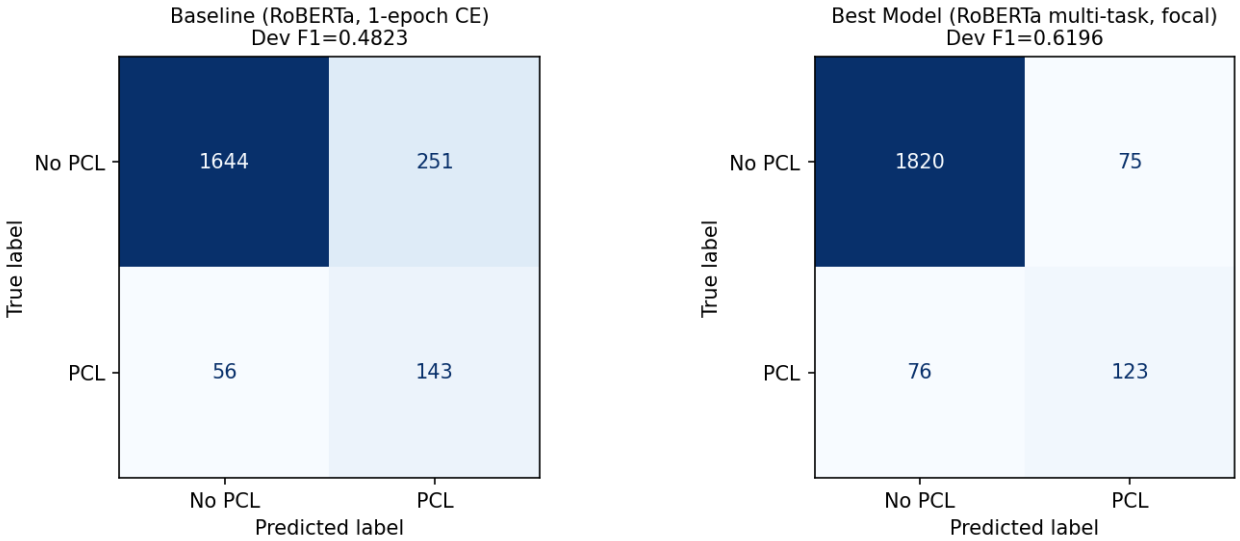

Figure 3: Confusion matrices for the baseline (left) and the best model (right) on the dev set. The best model substantially increases true positives (74 vs. 44 at the same scale) while keeping false positives comparable.

Only best model correct — qualitative examples. The 199 cases where the best model succeeds but the baseline fails are dominated by examples annotated with *Presupposition*, *Compassion*, and *Authority voice* categories:

- *"Real poverty of Britain: Shocking images of UK in the Sixties where poor really meant poor"* — gold=1, categories: Authority voice + Compassion. The framing "poor really meant poor" presupposes a hierarchy of deprivation, a nuanced stance visible to the multi-task encoder (trained to detect *Authority voice*) but not to the single-task baseline.
- *"Government support to bring the Housing First programme to Wellington will make a real difference for the homeless"* — gold=1, categories: Unbalanced power relations + Presupposition. The presupposition that homeless people need external government action to "make a real difference" reflects an Unbalanced power relation that the richer multi-task representations detect.
- *"Hollywood star Leo Di Caprio urges help for reuniting immigrant children with their families"* — gold=1, category: Unbalanced power relations. The construction "[celebrity] urges help for [vulnerable group]" positions the community as passive objects of advocacy, a structural pattern captured by the category-aware encoder.

These examples suggest that the auxiliary category head, by forcing the shared encoder to represent fine-grained PCL mechanisms during training, improves detection of subtle framing patterns that a single-task model trained on limited data misses.

Only baseline correct / both wrong — failure modes. The 43 regressions are mostly positive examples where the best model under-predicts:

- *"In a 90-degree view of his constituency, one can see a high rise and a flyover while underneath it, homeless people sleep"* — gold=1, category: Unbalanced power relations. The juxtaposition (luxury infrastructure vs. sleeping rough) is an indirect rhetorical device; the model may focus on the neutral spatial description and miss the implicit power commentary.
- *"Our country is in need of serious change. We can continue to celebrate women's day, throw massive budgets at events..."* — gold=1, category: Shallow solution. The sarcastic dismissal of superficial gestures is a Shallow solution pattern that is difficult to detect without pragmatic inference.

The 108 *both wrong* cases are uniformly false negatives (gold=1 predicted as 0 by both models), indicating that these are genuinely hard PCL examples requiring reasoning beyond surface language. Many involve *Presupposition* (e.g. "Many refugees don't want to be resettled anywhere, let alone in the US") where PCL is embedded in a clause-level assumption rather than explicit condescending language.

Part B: Per-Keyword Analysis

To diagnose where the model performs well or poorly across the ten vulnerable-community search keywords used to construct the corpus, precision, recall, and F1 were computed separately for each keyword's subset of the dev set.

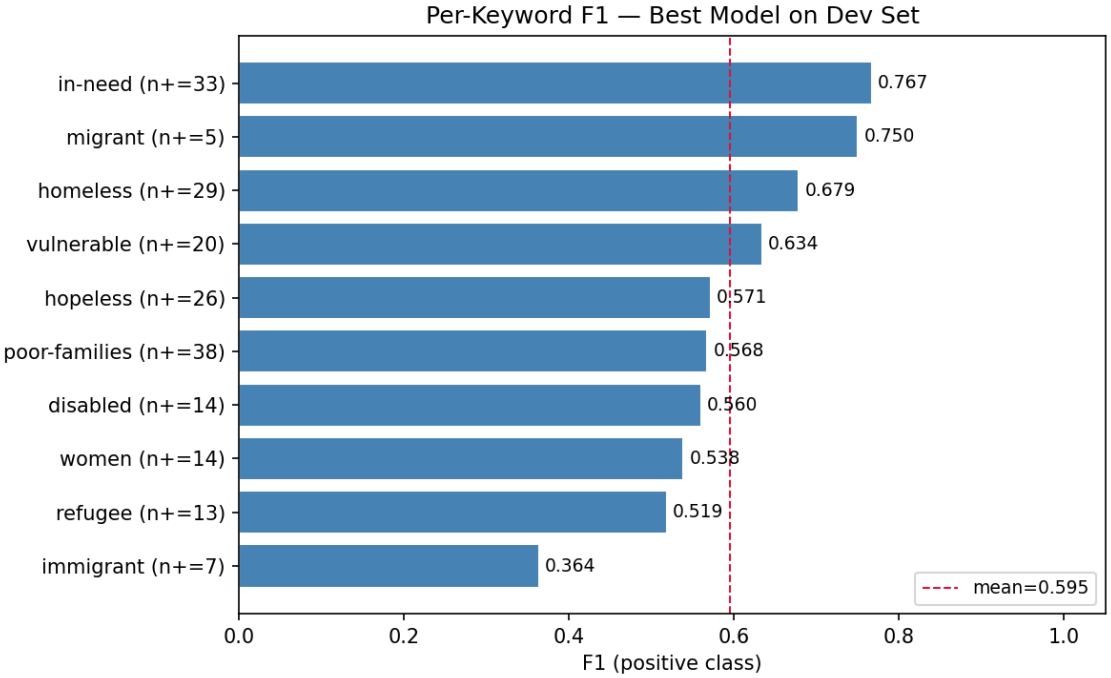

Figure 4: Per-keyword F1 of the best model on the dev set, sorted ascending. The dashed line shows the macro-average F1 across keywords.

The model performs best on *in-need* (F1 = 0.767) and *migrant* (F1 = 0.750). The *in-need* category likely benefits from having the highest positive count (33) and from the EDA finding that *needs-oriented* framing is a strong PCL signal: texts about communities described as "in need" tend to use relational, helper-stance language that the model detects reliably. The perfect precision for *migrant* (P = 1.0) suggests that when the model does predict PCL for this keyword it is always right, though the low positive count ($n+ = 5$) makes F1 estimates unstable.

The weakest performance is on *immigrant* (F1 = 0.364), which also has the fewest positive examples ($n+ = 7$).

Keyword	N+	N–	P	R	F1
in-need	33	193	0.700	0.848	0.767
migrant	5	202	1.000	0.600	0.750
homeless	29	183	0.704	0.655	0.679
vulnerable	20	189	0.619	0.650	0.634
hopeless	26	191	0.533	0.615	0.571
disabled	14	180	0.636	0.500	0.560
poor-families	38	152	0.583	0.553	0.568
women	14	219	0.583	0.500	0.538
refugee	13	175	0.500	0.538	0.519
immigrant	7	211	0.500	0.286	0.364

Table 3: Per-keyword precision, recall, and F1 (positive class) of the best model on the dev set. N+ and N– are the number of positive and negative examples.

With so few positive training and evaluation examples, even a small number of missed detections dramatically reduces F1. This suggests the model has not learned robust PCL representations for the *immigrant* context, possibly because its PCL articles mix reporting styles or involve more politically charged (rather than purely condescending) language. *Refugee* (F1 = 0.519) and *women* (F1 = 0.538) also underperform, likely for similar reasons: PCL about women in particular can involve implicit presupposition and subtle authority-voice patterns that are harder to detect from surface cues alone.